

Deep Learning Assignment 1

1) Network details (architecture, parameters, training setting, rationale)

Task

- **Multi-task CNN:** (a) categorical facial expression classification (8 classes), and (b) continuous **valence/arousal** regression in $[-1,1]$.

Data interface

- Images at Dataset/images/ named as integers (0.jpg/png, ...).
- Annotations at Dataset/annotations/
 - k_exp.npy → integer class id.
 - k_val.npy, k_aro.npy → floats (valence/arousal).
 - k_lnd.npy (optional) → facial **68×2** landmarks; code accepts 136 flat or (2,68) and reshapes to (68,2). Missing landmarks handled with NaNs.
- A fast path also supports consolidated files (exp.npy, val.npy, aro.npy, lnd.npy).

Dataset class

AffectFaceDataset

- Converts to **RGB**, resizes to **224×224**, normalizes using ImageNet mean/std.
- Returns:
 - image: Tensor (3, 224, 224)
 - targets: {"exp": Long, "val": Float, "aro": Float}

Splitting & class imbalance

- **StratifiedShuffleSplit** on the expression label to preserve class distribution.
- Ratio: **70% train / 15% val / 15% test** (two-stage 70/30 then 50/50 of the 30).
- **Class weights** for CE are computed from train distribution:

$$w_c = \frac{\sum_i n_i}{C \cdot n_c}$$

and passed to CrossEntropy to counter imbalance.

Data augmentation (train only)

- RandomHorizontalFlip(p=0.5), RandomRotation($\pm 10^\circ$), ColorJitter(0.2,0.2,0.2,0.1), RandomAffine(translate=10%, scale 0.9–1.1), then normalize.
- **Rationale**: robust to pose/illumination/scale; reduces overfitting.
- **Val/Test**: deterministic resize + normalize only.

Backbones (transfer learning)

- **ResNet18** and **EfficientNet-B0**, pretrained on ImageNet when available.
- Classifier layers are **stripped**:
 - ResNet18: replace fc with Identity; feature dim = 512.
 - EfficientNet-B0: replace classifier with Identity; feature dim = 1280.
- The shared feature vector is fed to a custom **multi-task head**.

Multi-task head

Expression head: Linear(in_features $\rightarrow$ num_classes=8)

VA head: Linear(in_features $\rightarrow$ 128) $\rightarrow$ ReLU $\rightarrow$ Linear(128 $\rightarrow$ 2) $\rightarrow$ Tanh # [-1,1]

Outputs: {"expression": logits, "valence": v, "arousal": a}

Tanh enforces the target range for VA.

Losses (with missing-label masking)

$$\mathcal{L} = \alpha \cdot \text{CE}(\hat{y}, y) + \beta \cdot \text{MSE}(\hat{v}, v) + \gamma \cdot \text{MSE}(\hat{a}, a)$$

- Samples with missing VA labels use **sentinel -2** and are **masked out**.

Optimization & schedule

- **Adam** (or AdamW) with LR 3×10^{-4} , weight decay 1×10^{-4} .
- **ReduceLROnPlateau** on **val loss** (factor 0.5, patience 3).
- Epochs: **10** in the supplied run.
- **Best-on-val** state reloaded before test metrics.

Monitoring & utilities

- Per-epoch history: train_loss/acc, val_loss/acc, val_rmse, aro_rmse.
- Post-training metrics include **Accuracy**, **Macro-F1**, **RMSE_{val, aro}**, **Pearson r_{val, aro}**, **SAGR_{val, aro}**, **CCC_{val, aro}**.
- Confusion matrix & qualitative grid helpers are included (require a model object in memory).

Rationale for these baselines

- **ResNet18**: strong residual baseline with good gradient flow; small enough for fast iteration.
- **EfficientNet-B0**: parameter-efficient with good accuracy/FLOP trade-off.

2) Dataset splits

- Implemented via **StratifiedShuffleSplit** to maintain label proportions.
- Printout example already present: Split sizes: train=70, val=15, test=15
- Keep a fixed `random_state` for reproducibility.

3) Training graphs

- Code produces three panels per model: **Loss (train/val)**, **Accuracy (train/val)**, **RMSE (valence/arousal)**.
- Your screenshot shows monotonic loss decrease and stabilizing validation curves for both **ResNet18** and **Eff-B0**.

4) Performance measures

Categorical (expressions)

model	acc	f1_macro
resnet18	0.4633	0.4617
efficientnet_b0	0.4233	0.4156

Continuous (valence / arousal)

model	rmse_val	rmse_aro	corr_val	corr_aro	sagr_val	sagr_aro	ccc_val	ccc_aro
resnet18	0.3727	0.3312	0.615	0.509	0.767	0.773	0.556	0.444
efficientnet_b0	0.4012	0.3432	0.652	0.439	0.733	0.768	0.434	0.374

Which metric matters most “in the wild”?

For VA, prefer **CCC**: it rewards high correlation **and** agreement in mean/variance, which aligns with real-world consistency. For expressions on imbalanced sets, prioritize **macro-F1** over raw accuracy.

5) Performance comparison of CNN architectures (accuracy & time)

Accuracy

- **ResNet18** outperforms Eff-B0 on **acc** (+0.04) and **macro-F1** (+0.046), and dominates most VA metrics (lower RMSEs; higher SAGR/CCC; higher *r* on arousal).

- **Eff-B0** only wins **valence correlation** $r(+0.037)$, suggesting better linear correlation but poorer overall agreement (lower CCC).

Time

model	best acc	total time
resnet18	0.4633	2 hours
efficientnet_b0	0.4233	3 hours

Conclusion of comparison

Choose **ResNet18** as the primary model: better categorical performance and stronger VA agreement (CCC), with simpler optimization. Eff-B0 may be retained as a secondary model if maximizing **valence** *ris* specifically required.

6) Transfer learning details

- **Pretrained** ImageNet weights when available; automatic **fallback** to random init if download fails.
- Final classifier of the backbone is **removed**; features feed into custom heads.
- VA head ends with **Tanh** to respect the label domain $[-1,1]$.
- Class imbalance mitigated via **weighted CE**.
- LR is reduced on plateau of **val loss**.

7) Limitations & next steps

- **Gap train vs. val accuracy** → more regularization (label smoothing 0.05, Mild MixUp=0.1), or longer fine-tune with discriminative LR after unfreezing last stage.
- **Imbalance** → consider **class-balanced focal loss** or **WeightedRandomSampler**.
- **Calibration** → temperature scaling before ROC/PR analysis.
- Add **Kappa**, **Krippendorff's Alpha**, **macro ROC-AUC/PR-AUC** for the categorical head to fully match some rubrics.